

Day 17 Student Handout

Geometry Summer Credit | Central Angles, Inscribed Angles, and Arcs

Name:	Date:	Period:
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Learning Targets

I can identify central and inscribed angles and their intercepted arcs. I can find missing angle and arc measures. I can explain when to use equal, half, double, or 360-degree relationships.

Lesson 17A: Central Angles and Arcs

Warm-Up

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| 1. | Angles around a point are 95 degrees, 120 degrees, and x degrees. Find x . |
| 2. | A circle is divided into arcs of 80 degrees, 135 degrees, and x degrees. Find x . |
| 3. | In circle O , what does it mean for O to be the center? |

Central Angle Relationship

A central angle has its vertex at the center. Its measure equals the measure of its intercepted arc. All arcs around a circle total 360 degrees. A semicircle measures 180 degrees.

Problem	Work / notes
Central angle AOB measures 72 degrees. Find minor arc AB .	Vertex location: Relationship: Arc measure:
Minor arc CD measures 118 degrees. Find central angle COD .	Relationship: Angle measure:
Minor arc EF measures 145 degrees. Find major arc EF .	Whole circle equation: Major arc:

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Lesson 17B | Inscribed Angles and Intercepted Arcs

Inscribed Angle Relationship

Half the Intercepted Arc

An inscribed angle has its vertex on the circle. Its measure is half the measure of its intercepted arc:
 $\text{angle} = (1/2)(\text{arc})$. Therefore $\text{arc} = 2(\text{angle})$. Inscribed angles intercepting the same arc are congruent.

Problem	Work / notes
Inscribed angle ACB intercepts an 86-degree arc AB. Find angle ACB.	Vertex location: Relationship: Angle measure:
Inscribed angle XYZ measures 57 degrees. Find intercepted arc XZ.	Relationship: Arc measure:
Inscribed angles ACB and ADB intercept the same arc AB. If angle ACB=38 degrees, find angle ADB.	Same arc idea: Angle measure:
An inscribed angle intercepts a diameter. Find the angle measure and explain.	Intercepted semicircle: Angle measure:

Decision Routine

Before Calculating

1. Locate the vertex: center or circle? 2. Mark the intercepted arc. 3. Choose equal for central angles or half/double for inscribed angles. 4. Use 360 degrees only when combining arcs around the full circle.

4.	A central angle and an inscribed angle intercept the same 124-degree arc. Find both angles.
5.	Explain why the central angle is twice the inscribed angle when they intercept the same arc.